

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL

Class: XI 'Science'

Time: 1:30 hrs.

PRE-TERM EXAMINATION 2082

MORNING SHIFT 'SET-A'

Subject: Chemistry [3011]

F.M: 37

P.M: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'A' [9x1=9]

MULTIPLE CHOICE QUESTIONS

- $H_2S + H_2O_2 \longrightarrow S + H_2O$ shows
 - Reducing action of H_2O_2
 - Oxidizing action of H_2O_2
 - Catalytic action of H_2O_2
 - Oxidizing action of H_2S
- Strongest oxidizing agent among the following is
 - HClO
 - HClO₂
 - HClO₃
 - HClO₄
- Oxidation state and valency of carbon in CH_2Cl_2 is
 - 0, 4
 - 0, 2
 - +4, 0
 - 4, 0
- Absolute zero is the temperature equal to
 - Room temperature
 - 273° C
 - 273.16° C
 - 273.16° C
- The curve plotted between P and V at constant temperature is called
 - Isotherm
 - Isobar
 - Isochore
 - Isotope
- Which non-metal is most active?
 - F₂
 - N₂
 - O₂
 - S
- Which of the following species has the highest electron affinity?
 - F⁻
 - O
 - O⁻
 - Na⁺
- The size of following species increase in the order
 - $Mg^{+2} < Na^+ < F^- < Al$
 - $F^- < Al < Na^+ < Mg^{+2}$
 - $Al < Mg^{+2} < F^- < Na^+$
 - $Na^+ < Al < F^- < Mg^{+2}$
- The density of NH_3 at 30° C and 5 atm pressure is
 - 2.30 gm/lit
 - 3.42 gm/lit
 - 23 gm/lit
 - 30 gm/lit

GROUP - 'B' [4X5=20]

SHORT ANSWER QUESTIONS

1. a) State Modern Periodic Law. [1]
b) Write the main features of Modern Periodic Table. [4]
2. a) Define the terms:
(i) Screening Effect (ii) Effective Nuclear Charge [1+1]
b) Define ionisation energy. Why is the ionisation energy of nitrogen higher than that of Oxygen? [1+2]
3. a) Derive combined gas equation. [2]
b) Find the value of R in C.G.S. unit. [2]
c) What is significance of R? [1]
4. a) Volume of moist gas at 23° C and 800 mm Hg pressure is 100 cc. Calculate the aqueous tension at 23° C if the volume occupied by gas at NTP is 94.53 cc. [3]
b) Write any two applications of Boyle's Law. [2]

GROUP - 'C' [1X8=8]

LONG ANSWER QUESTION

1. a) Differentiate between oxidation number and valency. [2]
b) Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent in the reaction,
$$KMnO_4 + KCl + H_2SO_4 \longrightarrow MnSO_4 + K_2SO_4 + H_2O + Cl_2$$
 [2]
c) Justify that, " SO_2 acts as both oxidizing agent as well as reducing agent". [2]
d) Calculate the oxidation number of underlined species. [2]
i) $\underline{Ni}$ (CO)₄ ii) $\underline{Cu}$ (NH₃)₄SO₄ iii) NH₄ $\underline{N}$ O₃ iv) $\underline{S}$ O₄²⁻

Best of Luck